Pick Momentum

Build a research-backed equity portfolio in 10 minutes a month.

App: pick-momentum.streamlit.app · Free · No login required · Github

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Most people who want to invest seriously run into the same wall. They know they should be doing something smarter than picking stocks based on news, tips, or gut feel — but they don't know where to start.

Pick Momentum is a free tool built for exactly that situation. It takes a decades-old, academically proven strategy — momentum investing — and makes it usable for any individual investor with a brokerage account and about 10 minutes at the end of each month.

No black box. No magic. The logic is visible, the signals are explainable, and you stay in full control.

1. What Is Momentum Investing?

Here's the core idea: stocks that have gone up the most over the past 6–12 months tend to keep going up over the next 1–6 months. Stocks that have fallen the most tend to keep falling. This isn't a hot take — it's one of the most replicated findings in all of finance.

The intuition makes sense once you think about how markets actually work. When a company reports great earnings, not every investor processes the news on day one. Information spreads gradually — through analyst reports, media, word of mouth. So the stock drifts upward over months, not in a single spike. Momentum investing says: notice that drift early, ride it, and exit when it fades.

This isn't trend-following based on vibes. In 1993, Narasimhan Jegadeesh and Sheridan Titman published the first rigorous proof in The Journal of Finance, studying 24 years of US stock data. They found that buying past winners and avoiding past losers earned roughly 1% per month — about 12% annualised — consistently. Their paper has since been cited over 10,000 times, and hundreds of follow-up studies have confirmed the effect in 40+ countries and across asset classes.

In India, studies on the Nifty 500 (2001–2023) found momentum returns of 1.5–2% per month — higher than the US, likely because Indian markets have more retail participation and less efficient price discovery. More on that in Section 6.

Pick Momentum gives you a systematic, rules-based way to run this strategy yourself.

2. The Four Strategies

The tool offers four momentum variants, each with a different risk-return trade-off. Here's the comparison at a glance:

Feature	Classic	Risk-Adjusted	Dual	52-Week High
Signal	12M past return	Return ÷ volatility	12M return + market filter	Nearness to 52W high
Bear protection	Weak	Moderate	Strong — exits to cash	Weak

Feature	Classic	Risk-Adjusted	Dual	52-Week High
Est. return (India)	18–22% p.a.	14–17% p.a.	16–20% p.a.	15–19% p.a.
Max drawdown	–45 to –55%	–20 to –30%	–20 to –30%	–35 to –50%
Sharpe Ratio	0.6–0.9	0.9–1.0	0.9–1.2	0.7–0.9
Complexity	Low	Medium	Low–Medium	Low
Best for	Aggressive growth	**Most investors ★**	Capital preservation	Mid/small-cap focus

★ Risk-Adjusted Momentum is the recommended starting point. Performance figures are estimates based on Indian academic studies — actual results will vary.

Classic Momentum

The original strategy. Rank every stock in the Nifty 500 by its 12-month return (skipping the most recent month to filter out short-term noise), pick the top 15, hold equally, rebalance monthly. Simple, transparent, high upside — but no safety net during crashes. Drawdowns of 40–55% are possible in bad markets. Best for investors with a long horizon and the stomach to ride out the rough patches.

Risk-Adjusted Momentum

A smarter version. Instead of ranking purely by return, it ranks by return divided by volatility. A stock that gained 20% with low volatility ranks above one that gained 25% but was all over the place. Barroso & Santa-Clara (2015) showed this one change nearly doubled the Sharpe Ratio and cut maximum drawdown from –91% to –26%. Best risk-return trade-off of the four. If you're not sure where to start, start here.

Dual Momentum

Has a built-in bear market switch. Each month, before picking stocks, it checks whether the Nifty 50's own 12-month return is positive or negative. If positive — pick the top 15 stocks as normal. If negative — move to cash. No equity exposure until the market recovers. Gary Antonacci's research (2012) showed this combination delivered ~17% annualised with a maximum drawdown of just –17.8%, vs –50.9% for buy-and-hold. The trade-off: you spend some time in cash during false alarms. Best for conservative investors.

52-Week High Momentum

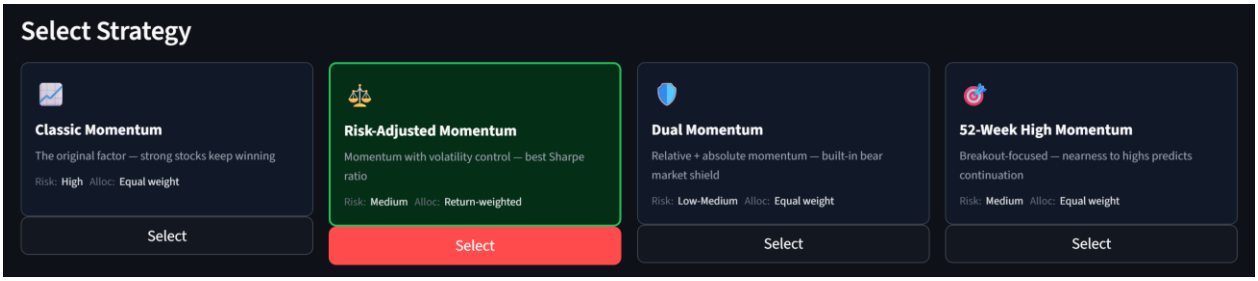
A different signal, same underlying idea. Ranks stocks by how close they are to their 52-week high price. The logic: investors psychologically anchor to the year-high and hesitate to buy at that level, causing the market to under-react when stocks approach it. When a stock finally breaks through, the subsequent move tends to be strong and sustained. George & Hwang (2004) found this signal generated returns beyond what classic momentum explained — and the effect is stronger in retail-driven markets like India.

3. Using the App: First-Time Setup

Open the app at pick-momentum.streamlit.app. No login. No data is stored on any server — your portfolio history lives entirely on your device.

Step 1 — Choose a Strategy

You'll see three tabs at the top: Portfolio Tool, Strategy Guide, and How to Use. Stay in Portfolio Tool. Four strategy cards appear. Click the green Select button on Risk-Adjusted Momentum to start. You can always switch strategies later.



Step 2 — Check the Market Regime

Below the strategy cards is the Market Regime panel. It tells you whether conditions are right to deploy capital:

- Green / UPTREND — Nifty 50 is above its 200-day moving average. Deploy your full intended capital.
- Red / DOWNTREND — Nifty 50 is below its 200-day moving average. Consider deploying 50% only, or wait.
- Red + CASH SIGNAL (Dual Momentum only) — Nifty 50's 12-month return is negative. Don't deploy. Move existing positions to a liquid fund.

Step 3 — Enter Your Capital and Run

Type the total rupee amount you want to invest across all 15 stocks. Then click Run Strategy. The app fetches the Nifty 500 constituent list from NSE India and downloads 15 months of price data. First run takes about 1–2 minutes. After that, data is cached for an hour — re-running within the same session is near-instant.

Step 4 — Review the Results

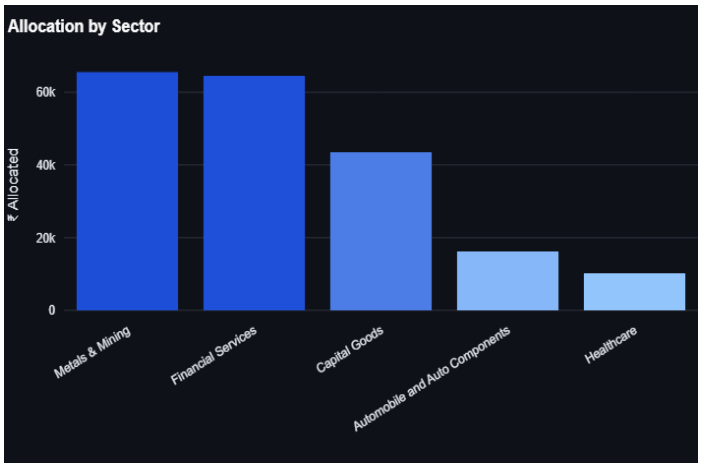
After the run, you get a full ranked table of 15 stocks with their momentum scores, allocated rupee amounts, and share counts. Below it, four charts: a score bar chart, a capital allocation donut, a return vs. volatility scatter plot, and a sector breakdown.

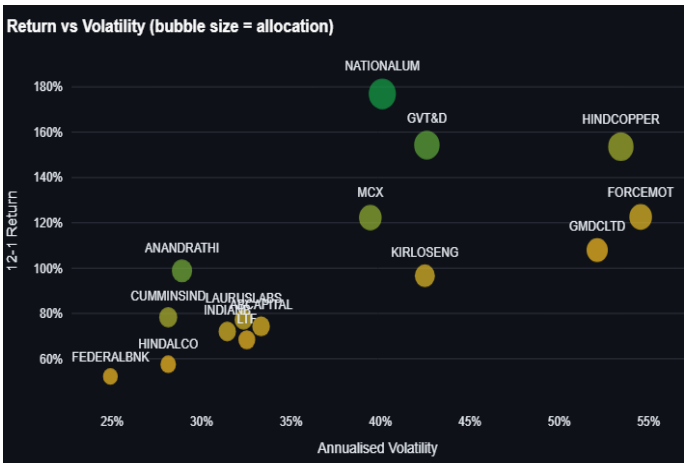
Top 15 Stocks — Risk-Adjusted Momentum

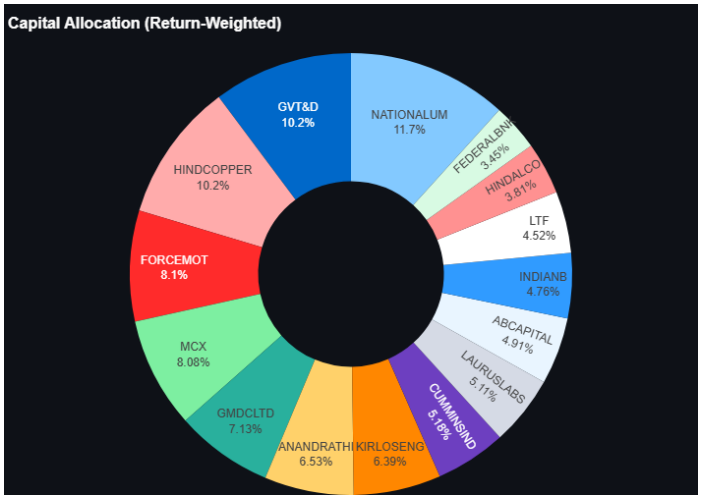
Selected	Avg Risk-Adj Score	Avg 12-1 Return	Avg Volatility
15	2.66	100.9%	37.6%

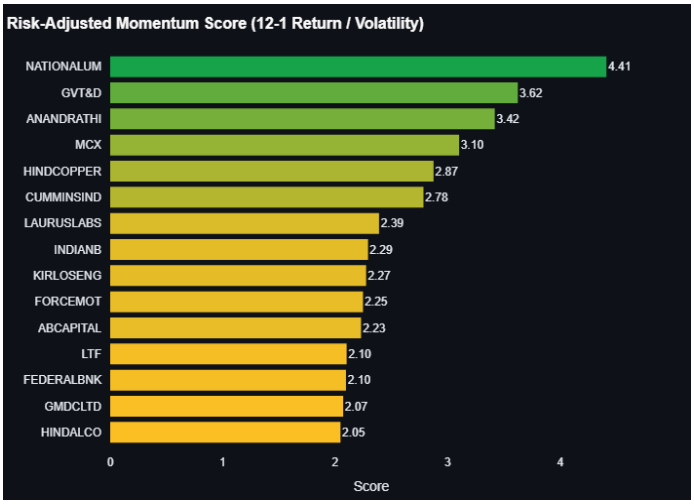
#	Action	Symbol	Company	Sector	Risk-Adj Score	12-1 Return	12M Return	Ann. Vol	vs 52W High	Weight	Allocated (₹)	Shares	CMP (₹)
1	NEW	NATIONALUM	National Aluminium Co. Ltd.	Metals & Mining	4.41	176.88%	171.25%	40.1%	-8.5%	11.69%	₹58,447	145	₹401.95
2	NEW	GVT&D	GE Vernova T&D India Ltd.	Capital Goods	3.62	154.33%	200.77%	42.6%	-3.0%	10.20%	₹50,995	11	₹4,625.50
3	NEW	ANANDRATHI	Anand Rathi Wealth Ltd.	Financial Services	3.42	98.80%	111.10%	28.9%	-2.2%	6.53%	₹32,645	9	₹3,608.80
4	NEW	MCX	Multi Commodity Exchange of India Ltd.	Financial Services	3.10	122.30%	159.14%	39.4%	0.0%	8.08%	₹40,413	13	₹3,097.80
5	NEW	HINDCOPPER	Hindustan Copper Ltd.	Metals & Mining	2.87	153.62%	176.40%	53.4%	-25.0%	10.15%	₹50,761	89	₹569.20
6	NEW	CUMMINSIND	Cummins India Ltd.	Capital Goods	2.78	78.34%	96.28%	28.1%	-0.0%	5.18%	₹25,886	4	₹5,401.00
7	NEW	LAURUSLABS	Laurus Labs Ltd.	Healthcare	2.39	77.29%	101.15%	32.3%	0.0%	5.11%	₹25,539	20	₹1,228.30
8	NEW	INDIANB	Indian Bank	Financial Services	2.29	72.04%	58.11%	31.4%	-12.7%	4.76%	₹23,805	27	₹865.55
9	NEW	KIRLOSENG	Kirloskar Oil Eng Ltd.	Capital Goods	2.27	96.64%	133.20%	42.5%	-0.5%	6.39%	₹31,933	18	₹1,728.90
10	NEW	FORCEMOT	Force Motors Ltd.	Automobile and Auto	2.25	122.58%	109.62%	54.6%	-19.0%	8.10%	₹40,504	1	₹20,877.00

The return vs. volatility scatter is the most useful chart for understanding your portfolio. Stocks in the top-left quadrant — high return, low volatility — are the highest-quality momentum names. The bigger the bubble, the more capital allocated to that stock.









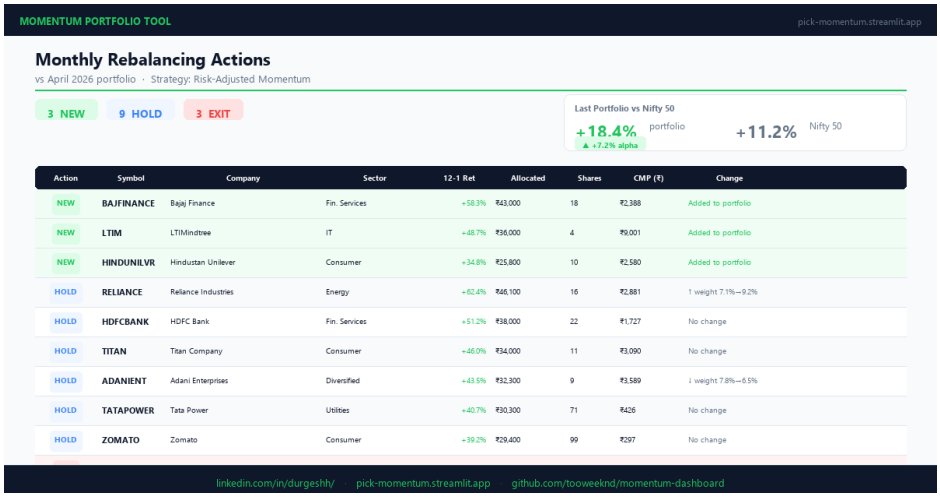
Step 5 — Save Your Run

Click Save Run & Download History at the bottom of the results. Save the file (momentum_history.json) to your device. This is your portfolio memory — without it, next month you won't see which stocks are NEW, HOLD, or EXIT, and you won't see performance vs. the index. Back it up to Google Drive or email it to yourself.

4. The Monthly Workflow

Run this on the last trading day of each month, after 3:30 PM IST. Running after market close ensures prices reflect the full day's move.

1. **Upload last month's momentum_history.json file.**
2. **Check the Market Regime. Has the trend changed since last month?**
3. **Run the strategy — same strategy, same capital amount.**
4. **Review last month's performance. How did your 15 picks do vs. Nifty 50?**
5. **Check the Rebalancing Actions panel — what's NEW, HOLD, or EXIT.**
6. **Place orders at market open on the first trading day of the new month.**
7. **Save the new history file. Replace the old version with the new one.**



Label	What it means	What to do
NEW	Not in last month's portfolio	Buy at market open on the 1st trading day of the new month
HOLD	In last month's portfolio and stays in the top 15	Do nothing. Keep your existing position.
EXIT	In last month's portfolio but dropped out of top 15	Sell at market open on the 1st trading day of the new month

A typical month involves 3–6 changes (exits + new entries). In steady bull markets, turnover is lower — sometimes just 1–2 swaps.

Before placing orders, check the sector breakdown. If 5 or more of your 15 stocks are from the same sector, you're running a momentum bet and a sector bet simultaneously. That's not necessarily wrong — sectors can trend strongly for months — but it's important to be aware of.

5. Why Momentum Works Especially Well in India

Momentum is a global phenomenon, but Indian markets have specific structural features that make it a particularly good fit:

- High retail participation. Over 90% of daily trading involves retail investors, who are more prone to anchoring, herding, and slow reaction to news — the exact biases momentum exploits.
- Analyst coverage gaps. Indian mid and small-cap stocks have far fewer analysts following them. Information diffuses more slowly, creating longer-lasting price trends.
- Earnings autocorrelation. Companies that beat estimates in one quarter tend to beat again the next — a fundamental driver that underlies price momentum.
- FII flow cycles. Foreign institutional investors rotate into sectors for multiple consecutive quarters, creating strong and persistent sector momentum.
- Index inclusion effects. Stocks entering Nifty indices attract concentrated passive buying pressure, producing sharp sustained moves — particularly useful for 52-Week High Momentum.

Academic research backs this up. Sehgal & Jain (2011) found momentum returns of 1.5–2% per month in the Nifty 500 — materially above US levels. Liu et al. (2011) confirmed the 52-week high effect is strongest in markets with high retail participation — India qualifies on both counts.

6. What to Watch Out For

Momentum investing has real risks. Strong long-run returns don't mean every month or year is smooth.

Momentum Crashes Are Real

All four strategies can lose 30–60% in rapid market reversals. This typically happens when beaten-down stocks rebound sharply after a panic selloff — crushing anyone who was holding their recent winners. The 2008 crisis saw Indian momentum portfolios drop 55–65%. If you can't see your portfolio halve at some point and still hold the strategy, start with Dual Momentum or Risk-Adjusted Momentum, not Classic.

Transaction Costs Add Up

Monthly rebalancing with 5 stock changes means roughly 10 trades per month. Here's the round-trip cost structure for NSE equity delivery:

Component	Rate	Notes
Brokerage (discount broker)	0.01–0.05%	Flat-fee brokers may charge ₹20 per trade instead
STT (Securities Transaction Tax)	0.1% (sell)	Delivery equity — mandatory
Exchange + SEBI charges	~0.003%	Fixed, minimal
Stamp duty	0.015% (buy)	State-mandated
Approx. round-trip total	**0.25–0.35%**	Per stock bought and later sold

On a ₹5 lakh portfolio with 5 monthly changes, expect ₹4,500–6,300 in annual friction — roughly 1–1.3% of capital. Factor this into return expectations. Monthly rebalancing also generates

short-term capital gains (STCG, taxed at 20% in India). Hold winners for 12+ months where possible to shift gains to LTCG (10%).

The Behavioural Trap

The biggest risk isn't the market. It's you. Momentum asks you to do two things that feel unnatural: buy stocks that have already run up (feels expensive), and sell stocks that have already fallen (feels like cutting your losses too late). Most investors override these signals based on conviction or emotion — and systematically underperform the mechanical strategy as a result.

If a stock is flagged EXIT, sell it. If a stock is NEW, buy it. The systematic advantage of a rules-based strategy disappears the moment you start making ad-hoc exceptions. And don't switch strategies after a patch of underperformance — strategies rotate in and out of favour. Switching after a bad period often means you leave just before the recovery.

This tool does not constitute a SEBI-registered investment advisory service. All signals are for educational and informational purposes only. Past performance figures are based on academic research and simulations — actual results will differ. Consult a qualified financial advisor before making investment decisions.

References

All four strategies are grounded in peer-reviewed academic research. Full citations:

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